

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY AND EDUCATION

DDM501 - Final Project Report

End-to-End Fraud Detection System: From Risk Scoring to Decision Intelligence

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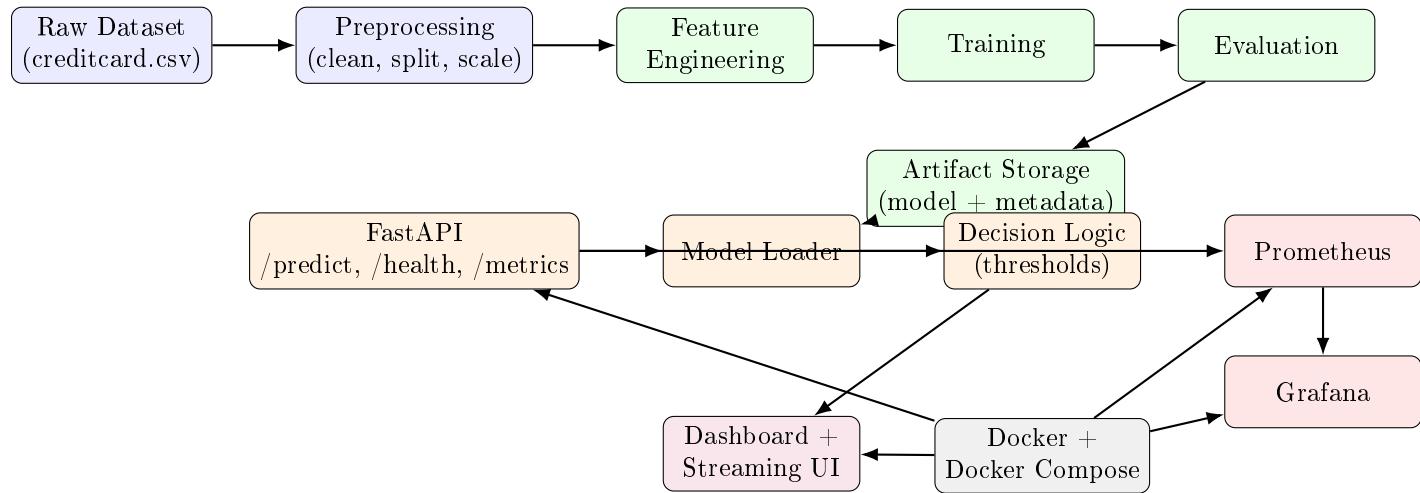
1 Executive Summary

This report presents a complete end-to-end Fraud Detection System aligned with DDM501 requirements, from data ingestion and model training to real-time API serving, dashboard-based decision support, and production-style monitoring. The core design principle is to separate *risk scoring* from *business decision policy*. The model output is treated as an **uncalibrated risk score**, then mapped by threshold policy to operational actions (allow, review, block).

The implementation includes a reproducible ML workflow, FastAPI serving layer, Dockerized deployment, Prometheus/Grafana observability, and automated tests via GitHub Actions. The report also covers Responsible AI constraints, explainability, fairness limitations, and realistic production gaps.

2 System Architecture (Critical)

2.1 Logical Architecture Diagram



Transaction → API → Model → Risk Score → Decision → UI → Monitoring

Figure 1: Logical end-to-end architecture (not a Docker-only topology).

2.2 Component Responsibilities

Data Layer. Stores historical transactions and applies preprocessing (schema checks, missing-value checks, scaling, and train/validation/test partitioning).

ML Pipeline. Executes feature engineering, model training, evaluation, threshold policy generation, and writes deployable artifacts.

Serving Layer. FastAPI validates payloads, loads model artifacts, computes risk score, and applies threshold decision logic.

Frontend Layer. Dashboard and streaming UI display decisions in real time and support analyst triage.

Monitoring Layer. Prometheus scrapes runtime metrics; Grafana visualizes system health and fraud decision distributions.

Deployment Layer. Docker and Docker Compose package and orchestrate API, UI, monitoring, and tracking services.

2.3 Interactions and Data Flow

1. Offline training creates model artifacts and threshold metadata in `artifacts/models/`.
2. API startup loads `final_model.joblib` and `model_info.json`.
3. Each incoming transaction is transformed to a feature vector and scored.
4. Decision logic maps score to a tier and action.
5. UI consumes API outputs for risk monitoring and case handling.
6. Prometheus and Grafana observe latency, error rate, and decision traffic.

2.4 Trade-Off Analysis: Latency, Complexity, Scalability

Latency. Logistic Regression yields low inference latency and stable CPU cost, suitable for near-real-time decisions.

Complexity. A simpler model and explicit policy reduce debugging and compliance burden compared to a highly tuned black-box stack.

Scalability. Dockerized stateless API can scale horizontally, but current demo stack lacks autoscaling and distributed stream processing.

3 A. Problem Definition and Requirements (10%)

3.1 Business Problem

Financial fraud produces direct chargeback losses and indirect operational costs. Fraud detection must maximize fraud capture while controlling false positives that hurt customer experience and increase manual review load.

3.2 Use Cases

1. Real-time scoring of a single transaction through REST API.
2. Analyst triage using review/high-risk tiers in a dashboard.
3. Monitoring and alerting for service performance and failure conditions.
4. Reproducible retraining and controlled deployment of new model artifacts.

3.3 Success Metrics

Model metrics: PR-AUC, ROC-AUC, precision/recall/F1 at policy thresholds.

System metrics: p95 latency (target $\leq 0.5s$), 5xx error rate, request throughput.

Business metrics: review capacity utilization, fraud captured in top-K buckets, and investigation efficiency (precision in review/high tiers).

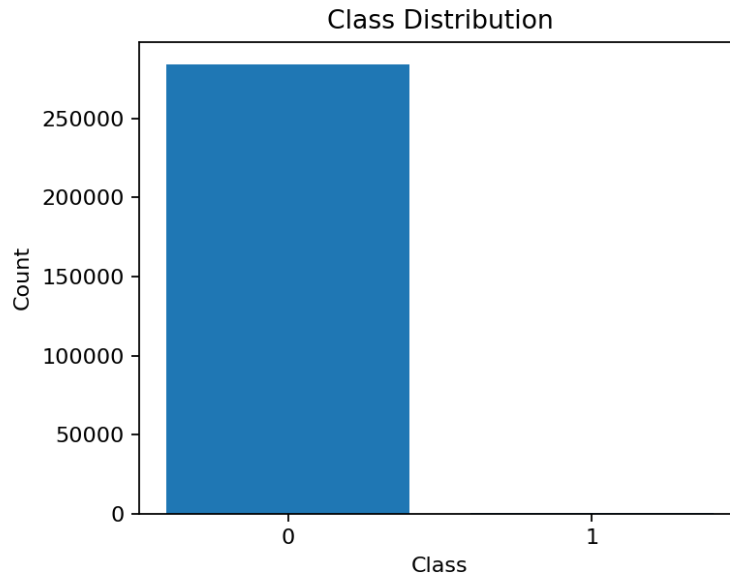


Figure 2: Class distribution of transactions, showing severe imbalance in fraud detection.

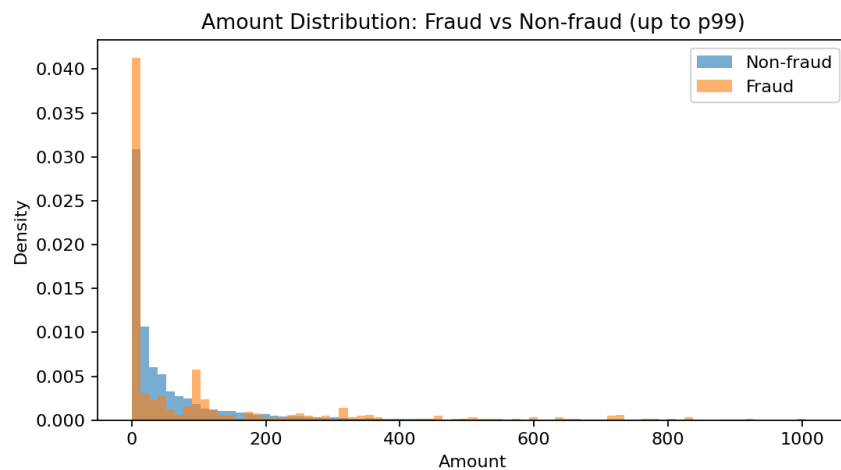


Figure 3: Amount distribution by class (up to p99), used to motivate robust threshold policy under heavy-tailed transaction values.

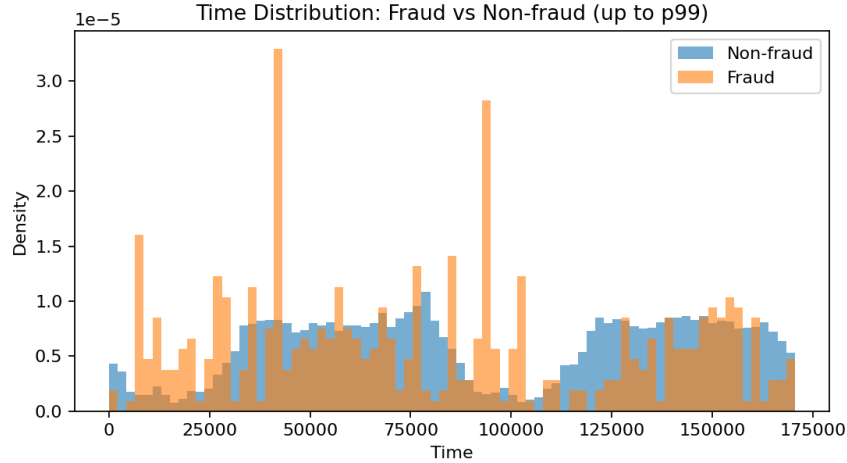


Figure 4: Time distribution by class (up to p99), highlighting temporal variation relevant for monitoring and drift analysis.

4 B. System Design and Architecture (15%)

Section 1 and its analysis satisfy the architecture requirement with a logical, component-driven model.

Technology stack justification:

1. **scikit-learn Logistic Regression:** robust baseline, interpretable coefficients, low operational complexity.
2. **FastAPI:** typed schemas, OpenAPI docs, low-friction deployment.
3. **Prometheus + Grafana:** standard observability for latency, errors, and prediction traffic.
4. **Docker Compose:** fast reproducible multi-service deployment for academic demo and CI validation.

5 C. Implementation (40%)

5.1 1) ML Pipeline

Data ingestion: from `data/archive/creditcard.csv`, with fallback synthetic mode for fast local runs.

Preprocessing: schema checks, feature selection, scaling pipeline, class imbalance handling.

Model training: baseline Logistic Regression and candidate LightGBM are benchmarked.

Evaluation: validation-first selection, threshold analysis, and final test metrics exported to `artifacts/models/model_info.json`.

Experiment tracking: MLflow is included in deployment stack for run and artifact tracking.

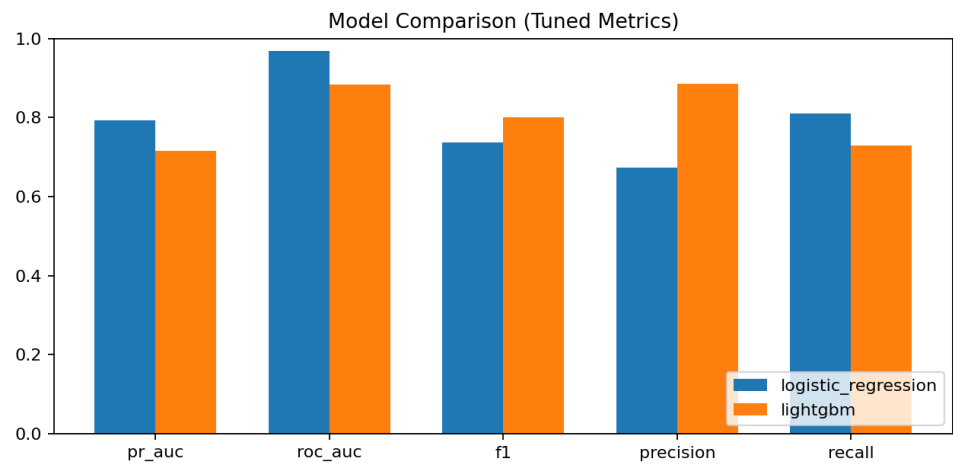


Figure 5: Tuned metric comparison between Logistic Regression and LightGBM.

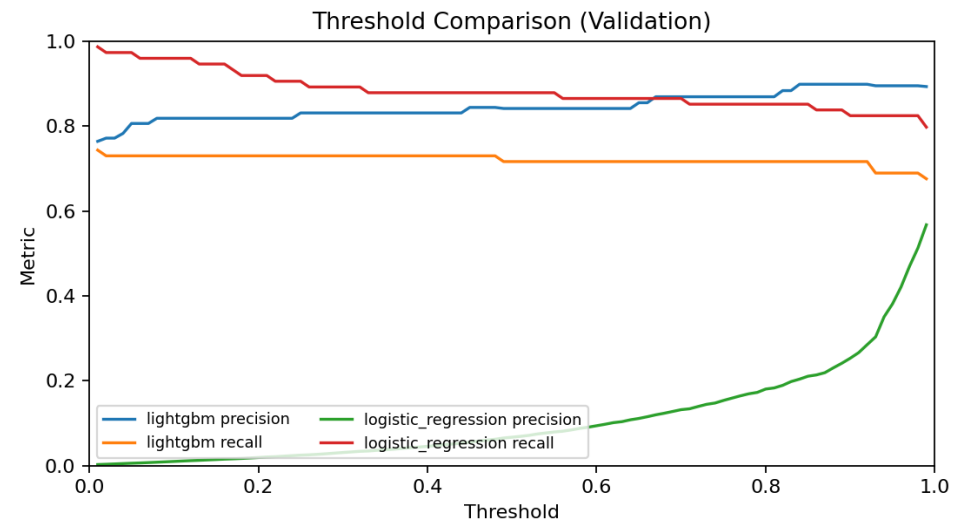


Figure 6: Validation threshold comparison used for model-policy analysis.

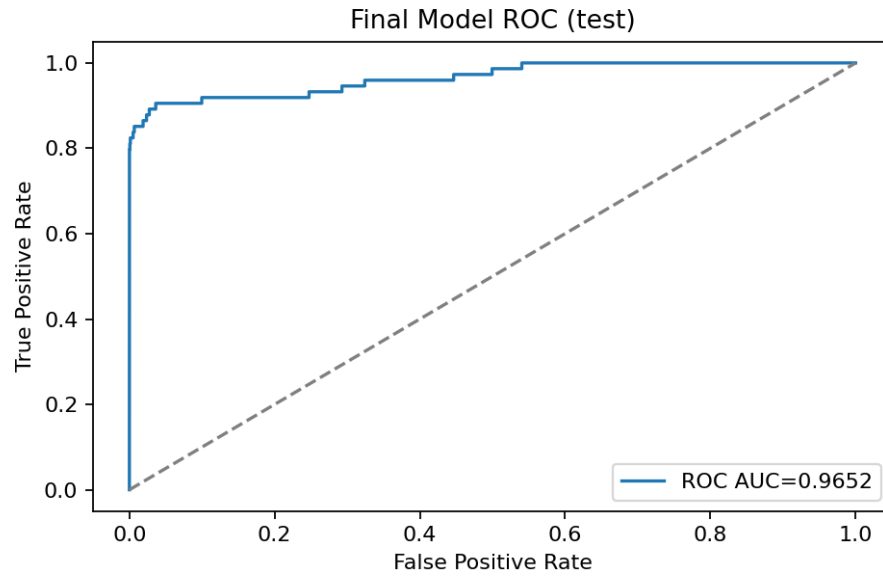


Figure 7: Final model ROC curve (test split).

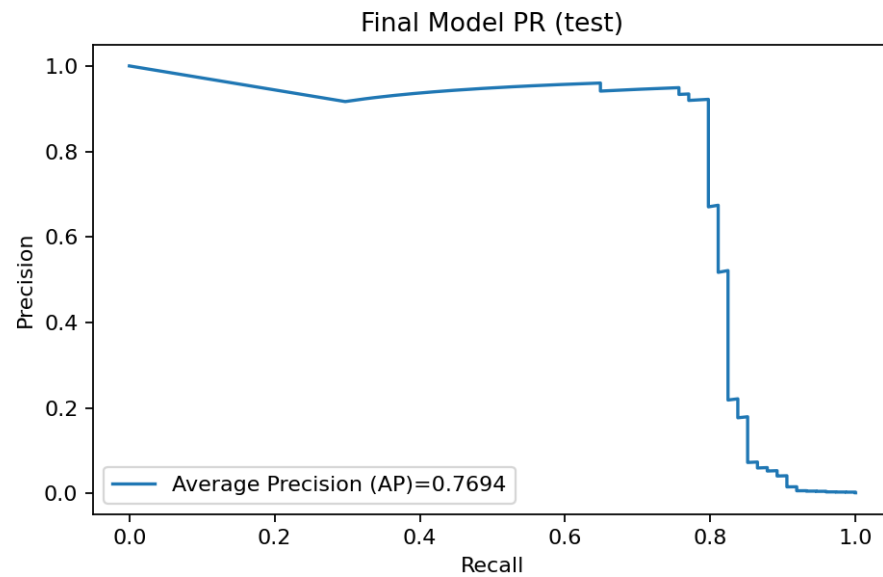


Figure 8: Final model PR curve (test split), emphasizing minority-class ranking quality.

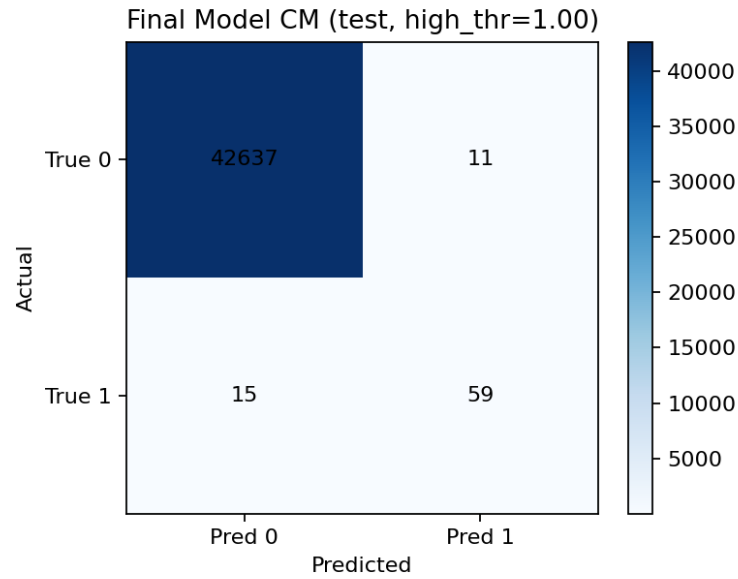


Figure 9: Final model confusion matrix at high threshold operating point.

5.2 2) Deployment

REST API: /predict, /health, /metrics, plus schema and stream-support endpoints.

Dockerfile: API and frontend are containerized under `deployment/api/Dockerfile` and `deployment/frontend/Dockerfile`.

Docker Compose: orchestrates API, frontend, Prometheus, Grafana, and MLflow services.

```

(.venv) and@andblap:~/Documents/Final_Project_DMM501_Group1$ docker compose -f deployment/docker-compose.yml up --build
=> transferring context: 2B
=> [mlflow internal] load .dockerignore
=> transferring context: 2B
=> [api internal] load .dockerignore
=> transferring context: 2B
=> [frontend 1/5] FROM docker.io/library/python:3.11-slim@sha256:233de06753d38d120b1a3ce359d8d3be8bda78524cd8f520c99883bfe33964cf
=> resolve docker.io/library/python:3.11-slim@sha256:233de06753d38d120b1a3ce359d8d3be8bda78524cd8f520c99883bfe33964cf
=> sha256:d98f36ba21593a4014243f41ec9c5370abd34f465a5d074e41bce23f56a165 14.37MB / 14.37MB
=> sha256:233de06753d38d120b1a3ce359d8d3be8bda78524cd8f520c99883bfe33964cf 10.37kB / 10.37kB
=> sha256:ade3166ed2838080a3f772e8f1ca733c77d631c0840ab1ca92311b0b350 1.75kB / 1.75kB
=> sha256:d1a45c0fb43d72e9c013ee97d460aa8f49f2c1bbd588b17a558cc6f7078b4276 5.48kB / 5.48kB
=> sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976 29.78MB / 29.78MB
=> sha256:37ffa6577b04e98598e2c10432eb8c66cf1af6c7698b21c6163f8334606aa14 1.29MB / 1.29MB
=> sha256:0816ee351b60c83c7a77a3f50c717891eed927f46a1d59f6cce5268065606 251B / 251B
=> extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976
=> extracting sha256:37ffa6577b04e98598e2c10432eb8c66cf1af6c7698b21c6163f8334606aa14
=> extracting sha256:d98f36ba21593a4014243f41ec9c5370abd34f465a5d074e41bce23f56a165
=> extracting sha256:0816ee351b60c83c7a77a3f50c717891eed927f46a1d59f6cce5268065606
=> [frontend internal] load build context
=> transferring context: 93.31kB
=> [api internal] load build context
=> transferring context: 223.21kB
=> [mlflow 2/5] WORKDIR /app
=> [mlflow 3/3] RUN pip install --no-cache-dir mlflow
=> [api 3/5] RUN apt-get update && apt-get install -y --no-install-recommends curl && rm -rf /var/lib/apt/lists/*
=> [frontend 4/5] COPY frontend /app/frontend
=> [api 4/7] COPY requirements.txt /app/requirements.txt
=> [api 5/7] RUN pip install --no-cache-dir -r /app/requirements.txt
=> [frontend 5/5] WORKDIR /app/frontend
=> [frontend] exporting to image
=> exporting layers
=> writing image sha256:b74f5f5bbc83cf95b1661705e8268b413cd4fd16e77af52fd8288c12eaf6a5d1
=> naming to docker.io/library/deployment-frontend
=> [frontend] resolving provenance for metadata file
=> [mlflow] exporting to image
=> exporting layers
=> writing image sha256:35a9cb4ab676632d15eb8f7de4cfe9a9a6c94069c91d58a2427062f5804c7c7
=> naming to docker.io/library/deployment-mlflow
=> [mlflow] resolving provenance for metadata file
=> [api 6/7] COPY src /app/src
=> [api 7/7] RUN mkdir -p /app/artifacts
=> [api] exporting to image
=> exporting layers
[+] up 35/35mg image sha256:b821f8336177e8a1f3e74c9d8f69c9ad70e6b8101d9185605c9c4b0ab5f6a2ea
✓ Image prom/prometheus:v2.54.1 Pulled
✓ Image grafana/grafana:11.1.0 Pulled
✓ Image deployment-mlflow Built
✓ Image deployment-api Built
✓ Image deployment-frontend Built
✓ Network deployment default Created
✓ Volume deployment mlflow data Created
✓ Volume deployment grafana data Created
✓ Container deployment-api-1 Created
✓ Container deployment-mlflow-1 Created
✓ Container deployment-frontend-1 Created
✓ Container deployment-prometheus-1 Created
✓ Container deployment-grafana-1 Created
Attaching to api-1, frontend-1, grafana-1, mlflow-1, prometheus-1

```

Figure 10: Docker Compose build and startup evidence for multi-service deployment.

5.3 3) Monitoring

Prometheus metrics: request count, latency histogram, prediction counters.

Grafana dashboard: API health and decision-distribution visualization.

Alert rules: high 5xx rate, high p95 latency, prediction rate anomalies.

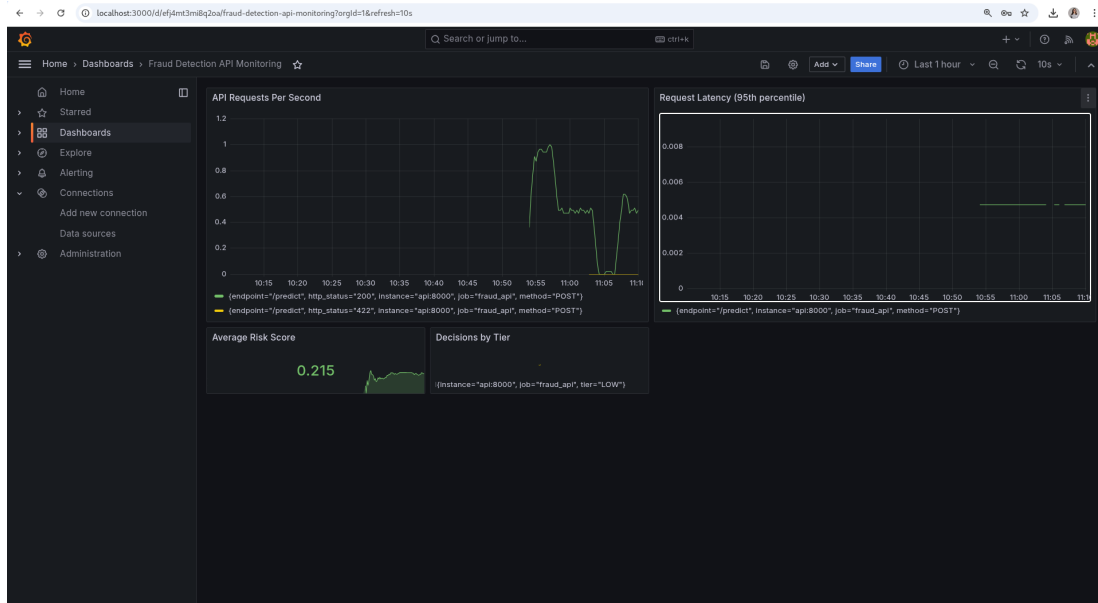


Figure 12: Grafana dashboard showing request rate, latency quantiles, and decision-tier traffic.

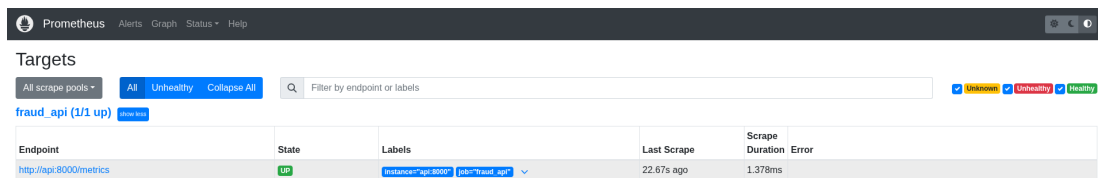
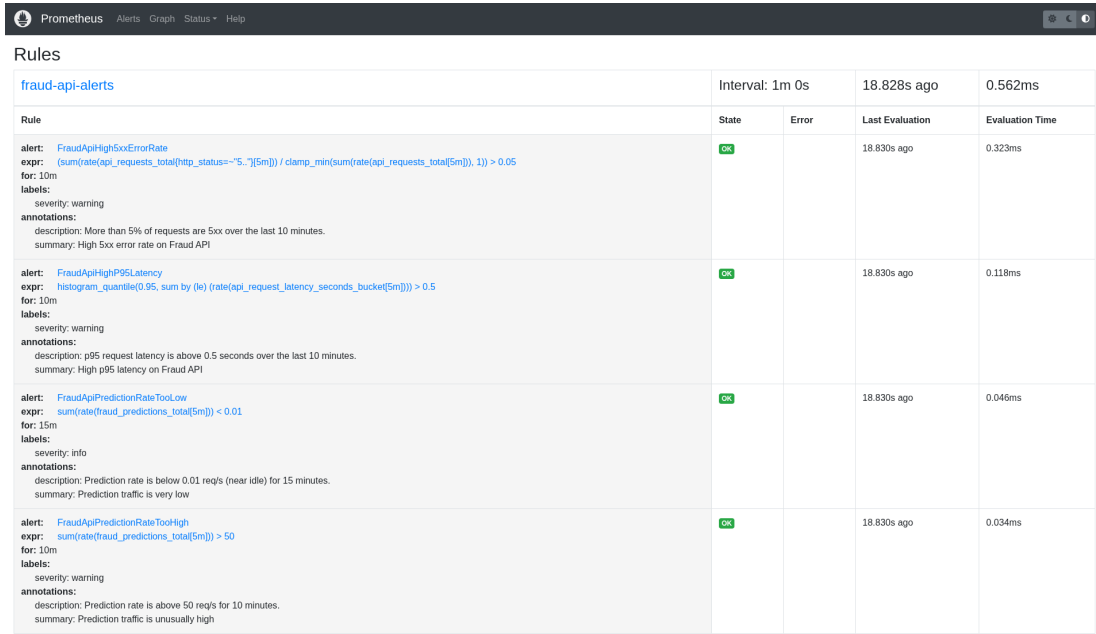


Figure 13: Prometheus targets confirming successful scrape of API metrics endpoint.



Rules			
Rule	Interval: 1m 0s	18.828s ago	0.562ms
Rule	State	Error	Last Evaluation
alert: <code>FraudApiHigh5xxErrorRate</code> expr: <code>(sum(rate(api_requests_total{http_status=~"5.."}[5m])) / clamp_min(sum(rate(api_requests_total[5m])), 1)) > 0.05</code> for: 10m labels: severity: warning annotations: description: More than 5% of requests are 5xx over the last 10 minutes. summary: High 5xx error rate on Fraud API	OK		18.830s ago
alert: <code>FraudApiHighP95Latency</code> expr: <code>histogram_quantile(0.95, sum by (le) (rate(api_request_latency_seconds_bucket[5m])) > 0.5</code> for: 10m labels: severity: warning annotations: description: p95 request latency is above 0.5 seconds over the last 10 minutes. summary: High p95 latency on Fraud API	OK		18.830s ago
alert: <code>FraudApiPredictionRateTooLow</code> expr: <code>sum(rate(fraud_predictions_total[5m])) < 0.01</code> for: 15m labels: severity: info annotations: description: Prediction rate is below 0.01 req/s (near idle) for 15 minutes. summary: Prediction traffic is very low	OK		18.830s ago
alert: <code>FraudApiPredictionRateTooHigh</code> expr: <code>sum(rate(fraud_predictions_total[5m])) > 50</code> for: 10m labels: severity: warning annotations: description: Prediction rate is above 50 req/s for 10 minutes. summary: Prediction traffic is unusually high	OK		18.830s ago

Figure 14: Prometheus alert rules status for latency, error ratio, and prediction-rate anomalies.

6 D. Testing and CI/CD (15%)

6.1 Test Coverage Types

1. **Unit tests:** preprocessing and utility functions.
2. **Integration tests:** API contract, health, docs, random features, and stream endpoints.
3. **Data quality tests:** placeholder checks exist and should be expanded to stronger schema/value guards.
4. **Model validation tests:** placeholder model-loading validation exists and should be extended for metric assertions.

6.2 GitHub Actions CI

CI runs on push and pull request, installs dependencies, executes pytest, and enforces coverage gate (80%). This supports minimum engineering rigor for DDM501.

7 E. Responsible AI (10%)

7.1 SHAP Explainability

SHAP summary artifacts are generated for global behavior analysis, helping reviewers inspect feature influence trends.

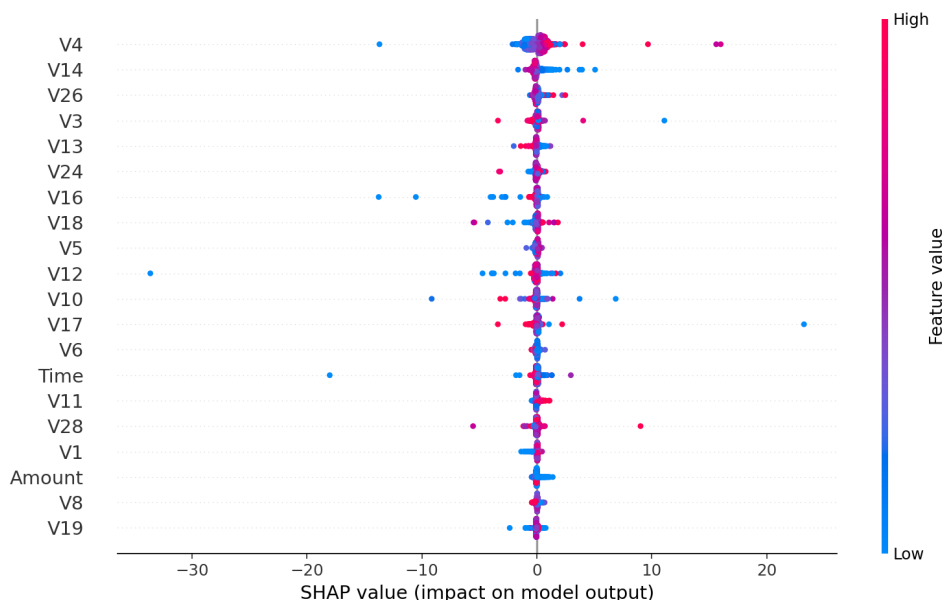


Figure 15: SHAP summary plot for global feature impact analysis (candidate model explainability evidence).

7.2 Fairness Limitation

Protected attributes are unavailable in the dataset; therefore demographic fairness claims are not possible. Only proxy-slice analysis (amount/time buckets) is feasible and should be interpreted with caution.

7.3 Ethics Discussion

Main ethical risks are false positives (customer friction), false negatives (fraud loss), and over-trust in model outputs. The system mitigates these by explicit threshold policy, human-in-the-loop review tier, and score semantic warnings.

8 F. Documentation (10%)

README: end-to-end project overview and requirements.

API docs: FastAPI OpenAPI and Swagger endpoint documentation.

Usage guide: Quick start instructions for training, API, frontend, and Docker stack.

9 Critical ML and Business Logic

9.1 1) Risk Score Is NOT Probability

The API returns `risk_score` with semantic tag `risk_score_uncalibrated`. It is a ranking signal, not a calibrated probability of fraud. Calibration can be distorted by class imbalance, class weighting, threshold policy optimization, and distribution shift.

9.2 2) Threshold Strategy: Top-K Policy

Current deployment metadata defines:

- `review_top_rate = 1%`
- `high_top_rate = 0.2%`

Why this policy:

1. Analyst capacity is finite, so review queue size must be constrained.
2. Fraud cost is asymmetric relative to review cost, so a high-risk tier supports stronger intervention at higher precision.
3. Top-K thresholds remain operationally robust when score calibration is imperfect.

9.3 3) Decision Layer Mapping

Decision logic is explicitly:

`risk_score → risk_tier → action`

With thresholds from metadata:

- if `score ≥ threshold_high` → HIGH → block
- else if `score ≥ threshold_review` → REVIEW → manual review
- else → LOW → allow

Table 1: Policy thresholds and test metrics from `artifacts/models/model_info.json`.

Operating point	Threshold	Precision	Recall	F1
Review tier (top 1%)	0.7391	0.1462	0.8514	0.2495
High tier (top 0.2%)	0.9999	0.8429	0.7973	0.8194
F1 comparator	0.9900	0.7024	0.7973	0.7468

10 Model Explanation: Logistic Regression vs LightGBM

10.1 Logistic Regression Details

For feature vector x , Logistic Regression computes:

$$\hat{s}(x) = \sigma(w^\top x + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Coefficients. Coefficients encode each standardized feature's signed contribution to log-odds. Magnitude helps rank global feature influence.

class_weight. Using `class_weight="balanced"` compensates for extreme class imbalance and improves minority-class sensitivity.

Regularization. L2 regularization stabilizes coefficients and reduces overfitting risk on sparse fraud signals.

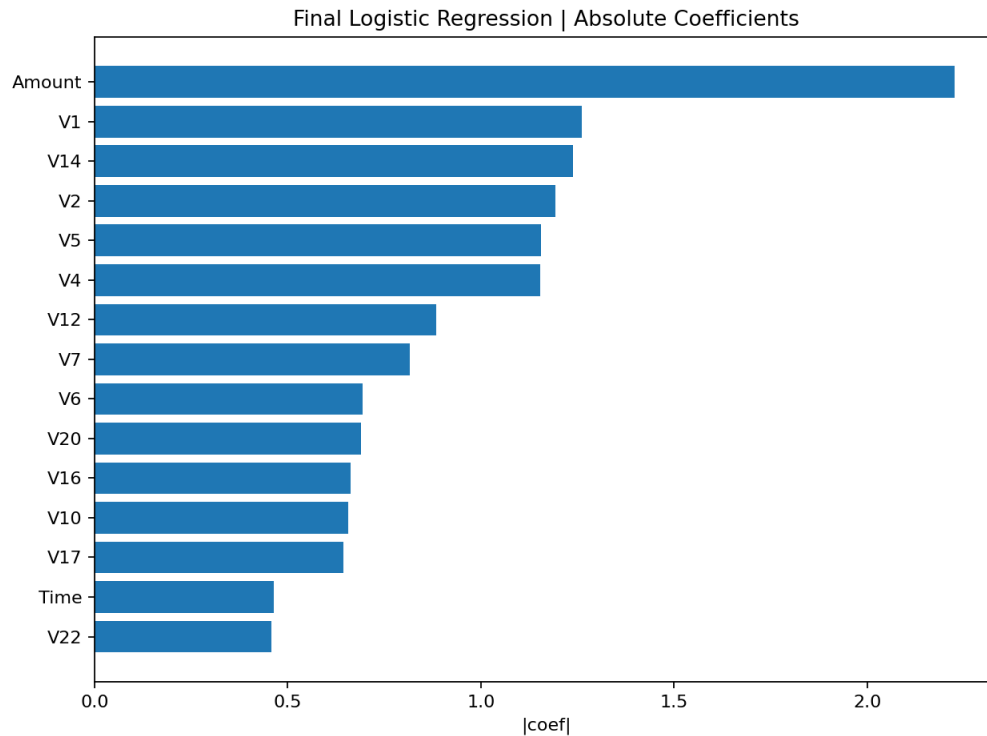


Figure 16: Absolute coefficients of the final Logistic Regression model (global interpretability view).

10.2 Why Selected over LightGBM

Based on stored validation results, Logistic Regression slightly outperformed the LightGBM candidate on validation PR-AUC while offering:

1. lower inference complexity,
2. easier governance/explainability,
3. more predictable behavior under constrained deployment.

Thus the project selected Logistic Regression for the production-like artifact set.

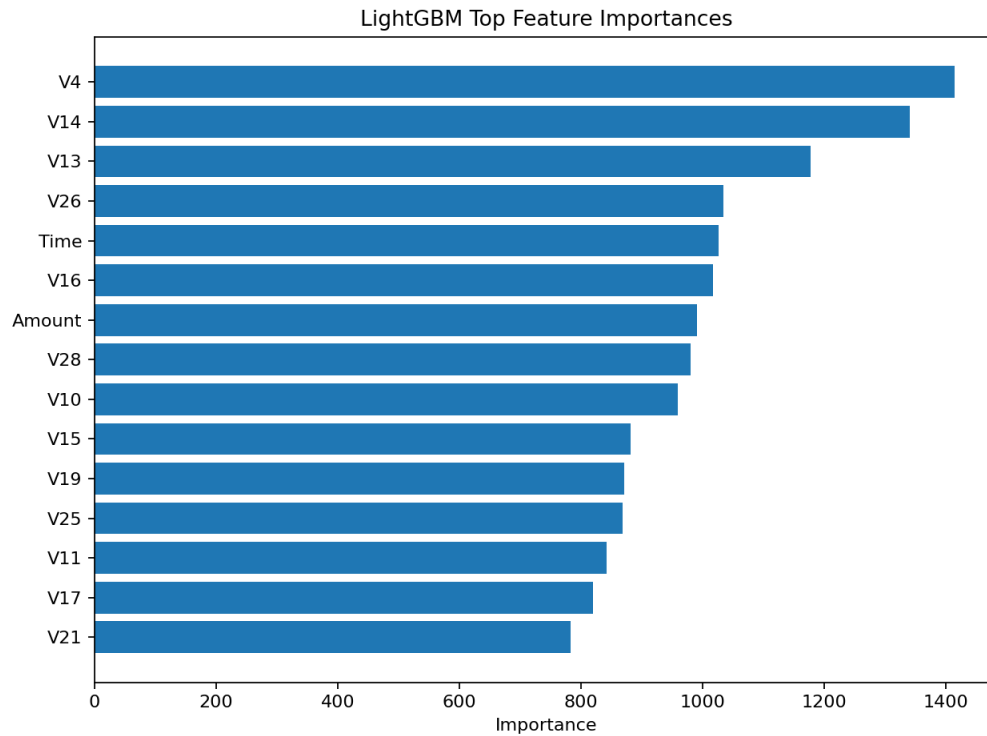


Figure 17: Top feature importances from LightGBM benchmark model (comparison reference).

11 System Deployment and Setup

11.1 Training Commands

```
python -m src.pipelines.run_model_workflow \
  --data-path data/archive/creditcard.csv \
  --artifacts-root artifacts \
  --seed 42
```

Fast synthetic run:

```
python -m src.pipelines.train_pipeline --data-path "" --artifacts-dir artifacts --fast
```

11.2 API Command

```
MODEL_PATH=artifacts/models/final_model.joblib \
MODEL_VERSION=creditcard-production-v1 \
uvicorn src.api.main:app --host 0.0.0.0 --port 8000
```

11.3 Frontend Command

```
cd frontend
python3 -m http.server 8082
```

11.4 Docker Compose Command

```
docker compose -f deployment/docker-compose.yml up --build
```

12 Frontend and Demo Workflow

12.1 Dashboard

The dashboard shows health, model metadata, threshold context, and risk-tier summaries to support daily fraud operations.

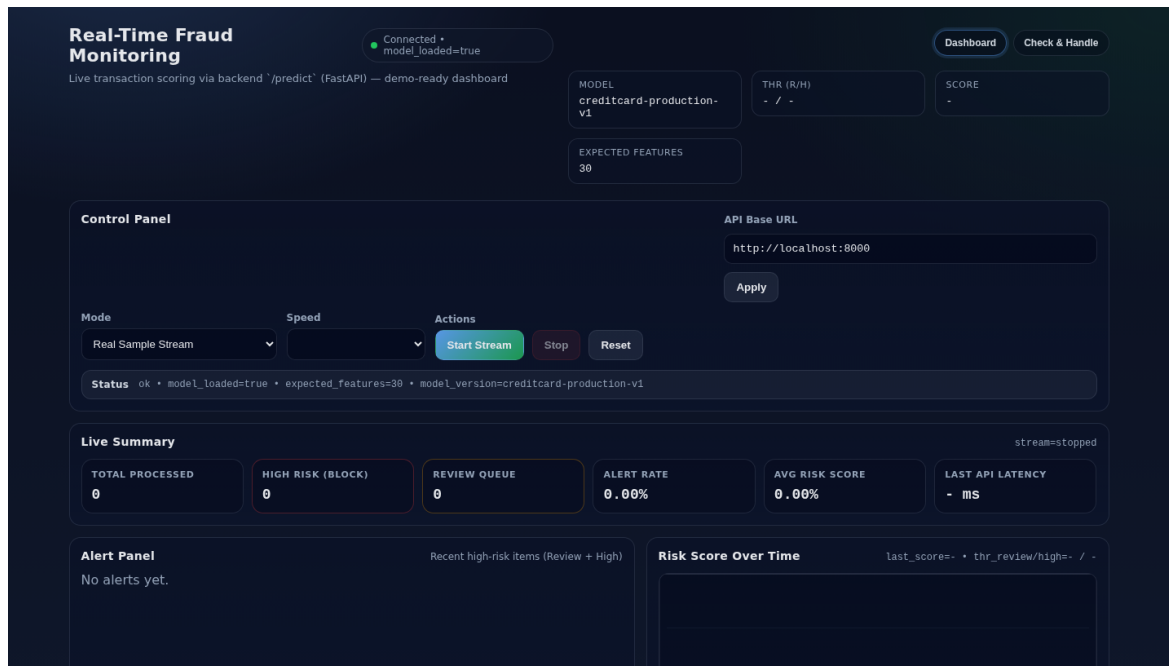


Figure 18: Frontend dashboard with streaming transactions, alert panel, and risk-over-time chart.

12.2 Real-Time Stream

A pull-based stream endpoint supplies scored events to the UI for live triage simulation, avoiding label leakage in public endpoints.

12.3 Alert System

UI-facing monitoring is backed by Prometheus alerts and Grafana dashboards, enabling operators to react to latency spikes, error bursts, or abnormal decision traffic.

12.4 Decision-Making Support

The UI surfaces risk tier and action as primary outputs. This design prevents over-interpretation of raw scores and keeps analysts focused on operational actions.

13 Limitations and Production Gap

1. **Simulation vs real system:** stream events are simulated, not from a production payment gateway.
2. **No calibration:** score is intentionally uncalibrated; no calibration curve/Brier monitoring in pipeline.
3. **No real streaming bus:** no Kafka-based ingestion, ordering guarantees, or exactly-once semantics.
4. **No closed feedback loop:** no automated label feedback, drift-triggered retraining, or rollout canary policy.

14 LaTeX Deliverable Completeness

This document satisfies the required LaTeX structure:

- **Document class:** `\documentclass[11pt]{article}`
- **Required packages included:** geometry, graphicx, booktabs, hyperref, amsmath, fancyhdr
- **Contains:** title page, table of contents, full sections, figures, and tables

15 Final Rubric Compliance Checklist

Table 2: DDM501 Rubric Compliance Status

Rubric Component	Status	Evidence Summary
Problem Definition (10%)	PASS	Business problem, use cases, and success metrics clearly defined.
Architecture (15%)	PASS	Logical architecture diagram + data flow + component responsibilities + trade-offs.
Implementation (40%)	PASS	ML pipeline, REST API, Docker deployment, monitoring stack implemented.
Testing and CI/CD (15%)	PARTIAL	Unit/integration + CI coverage gate present; data/model tests exist but depth can be expanded.
Responsible AI (10%)	PASS	SHAP artifacts, fairness limitations, and ethics discussion documented.
Documentation (10%)	PASS	README, API docs, and usage guide available and referenced.

15.1 Overall Result

Overall DDM501 Compliance: PASS (with noted testing-depth improvements).

16 Conclusion

The system delivers a full fraud decision pipeline from data to action. It is technically sound, business-aware, and production-aware for an academic deployment. The architecture is logically separated, the risk policy is explicit, and the report makes clear where demo constraints end and production engineering begins.